

Research Work – Understanding Fundamentals

Q.1

Difference between import module and from module import

Ans:

Difference between import module and from module import
import module loads the whole module, requiring
module.function(e.g.math.sqrt). from module import brings specific items
(e.g.from math import sqrt), letting you use sqrt directly. The first avoids
name clashes, the second saves typing but risks conflicts.

Q.2

Explain function parameters & return values

Ans:

Parameters are inputs a function takes (e.g., `a, b` in `add(a, b)`),
acting as placeholders. Return values are outputs it sends back (e.g., `a +
b`), usable after the function runs. They define what a function needs and
gives.

Q.3

What is variable scope (local vs global)?

Scope shows where a variable works. Local variables are inside
functions, only usable there. Global variables are outside, accessible
everywhere. Local avoids clashes; global needs `global` to change inside
functions.

Q.4

Why are built-in functions helpful? Name 5

They save time, are optimized, and handle common tasks. Examples:
`print()` (output), `len()` (length), `int()` (to integer), `range()` (sequence),
`type()` (data type).

Q.5

How do modules help in organizing code?

Modules split code into reusable files (e.g., `math.py`), grouping related functions. This reduces clutter, enables reuse, and simplifies collaboration by separating tasks.